

Freeze-and-add RBM: current status

Quick figure-led update, not a formal writeup. Toy model: only the pseudolikelihood curves, all scatter (data-vs-model) diagnostics, and the four-panel K -recovery heatmap. Pfam: every figure produced for the two reference runs – PF00072–PF00512 ($K = 50$, $N_{\text{iters}} = 10000$, $\text{paired_iters} = 5000$) and PF00072–PF01339 ($H_{\text{add}} = 20$), unfiltered. Conditional-generation results are not included here yet.

Toy model

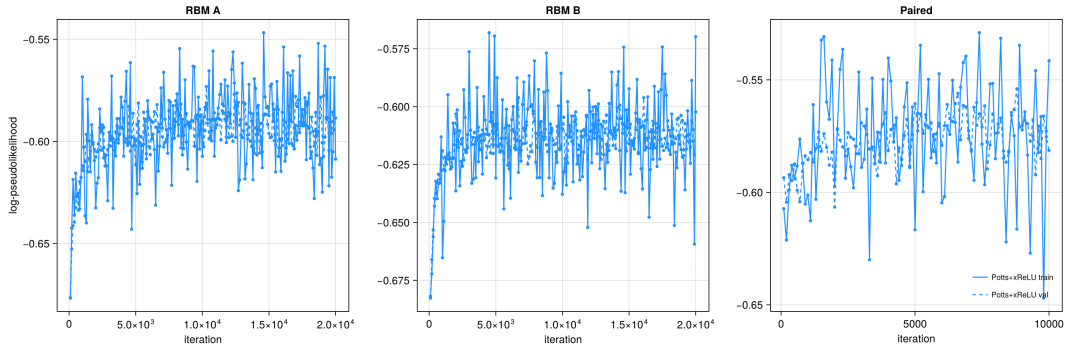


Figure 1: Pseudolikelihood, train/validation, private RBMs A , B , and paired model.

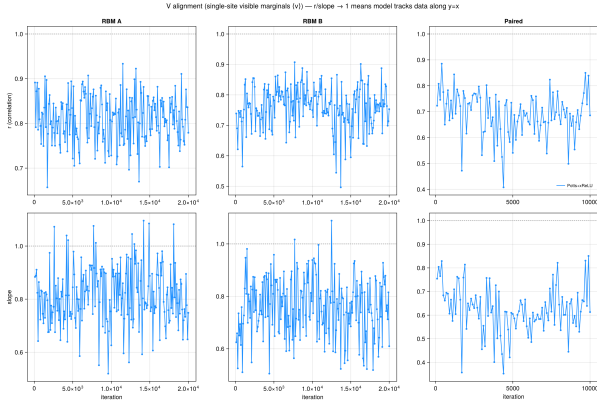


Figure 2: Single-site visible marginals $\langle v \rangle$, data vs. model.

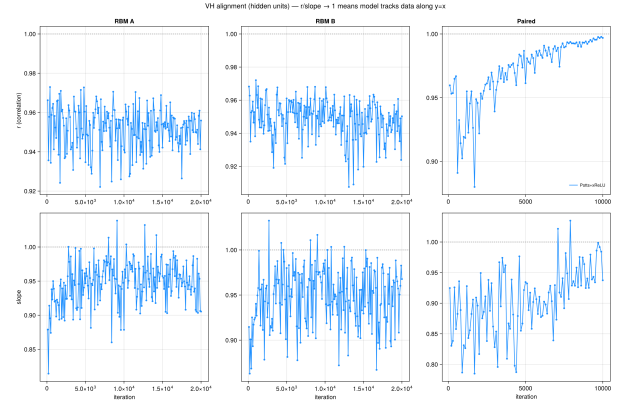


Figure 3: Joint $\langle vh \rangle$, added units, data vs. model.

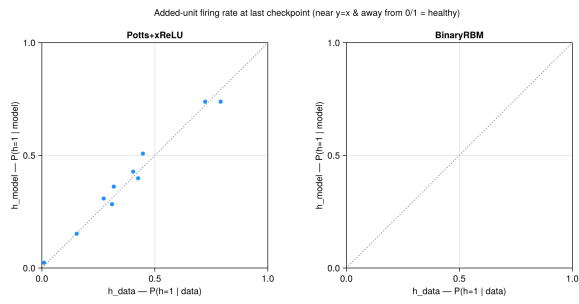


Figure 4: Added-unit firing rate, data vs. model (dead-unit check).

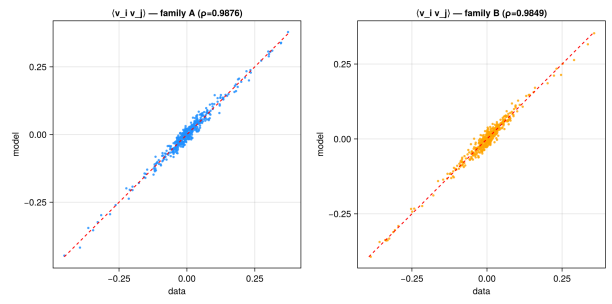


Figure 5: Private-RBM visible-visible correlation $\langle v_i v_j \rangle$ at equilibrium, families A/B .

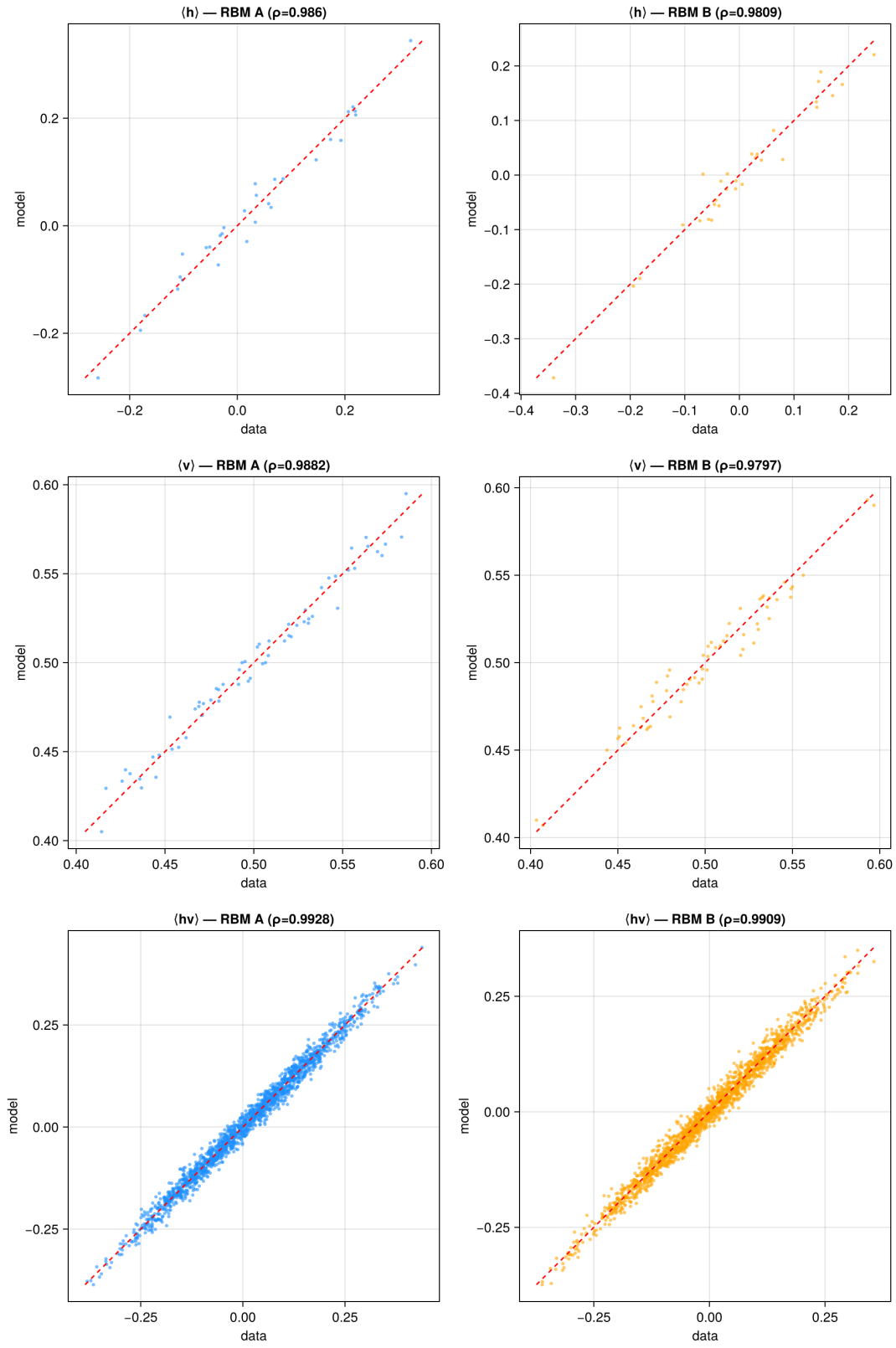


Figure 6: Private RBMs at equilibrium: $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$, data vs. model, families A (top) and B (bottom).

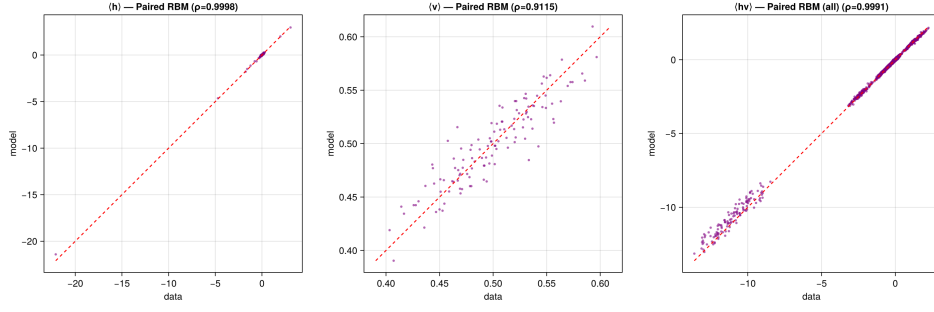


Figure 7: Paired model at equilibrium: $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$, data vs. model, all units.

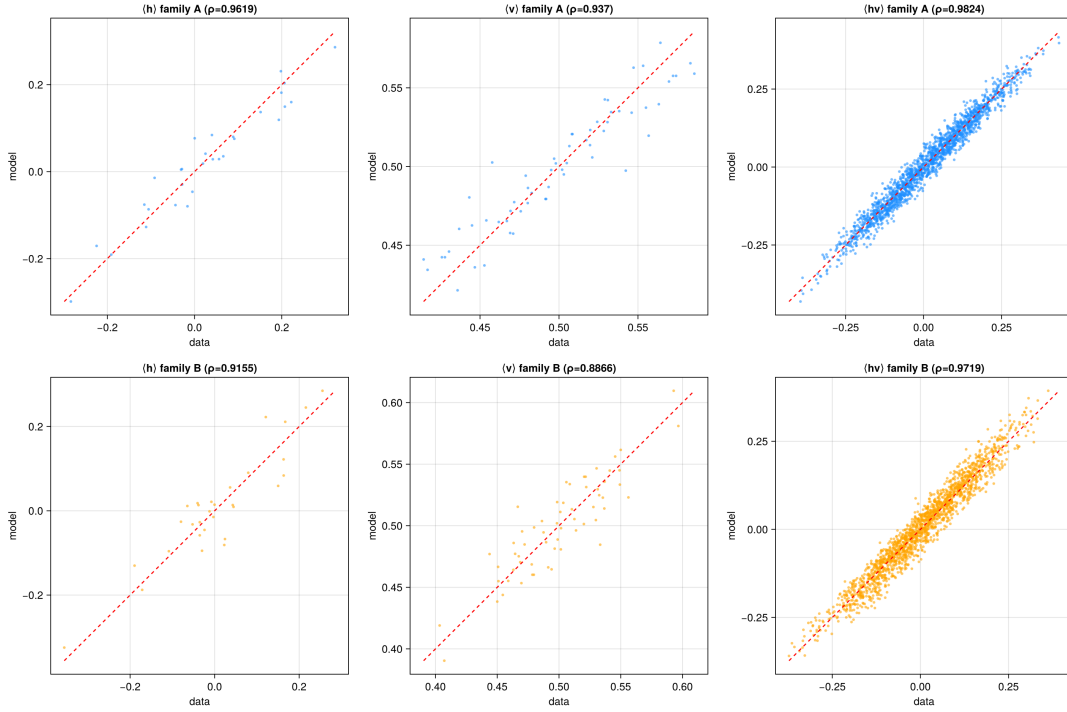


Figure 8: Intra-family $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$ at equilibrium, families A (top) / B (bottom).

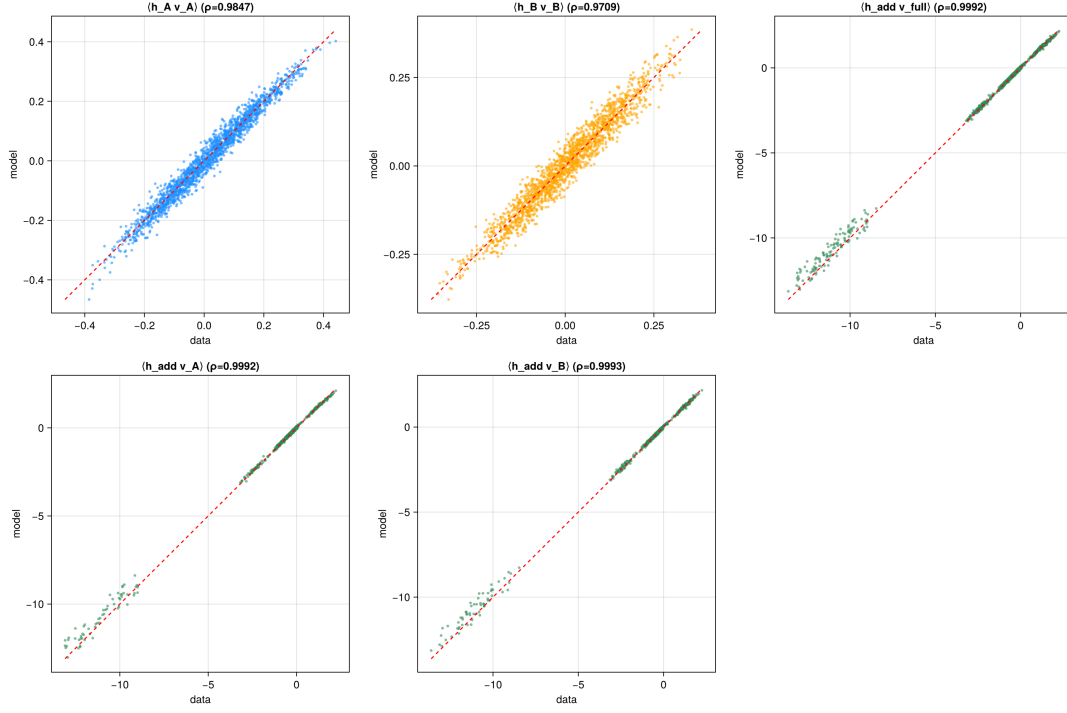


Figure 9: Block-wise $\langle hv \rangle$ at equilibrium: frozen A - A , frozen B - B , added units vs. full visible layer.

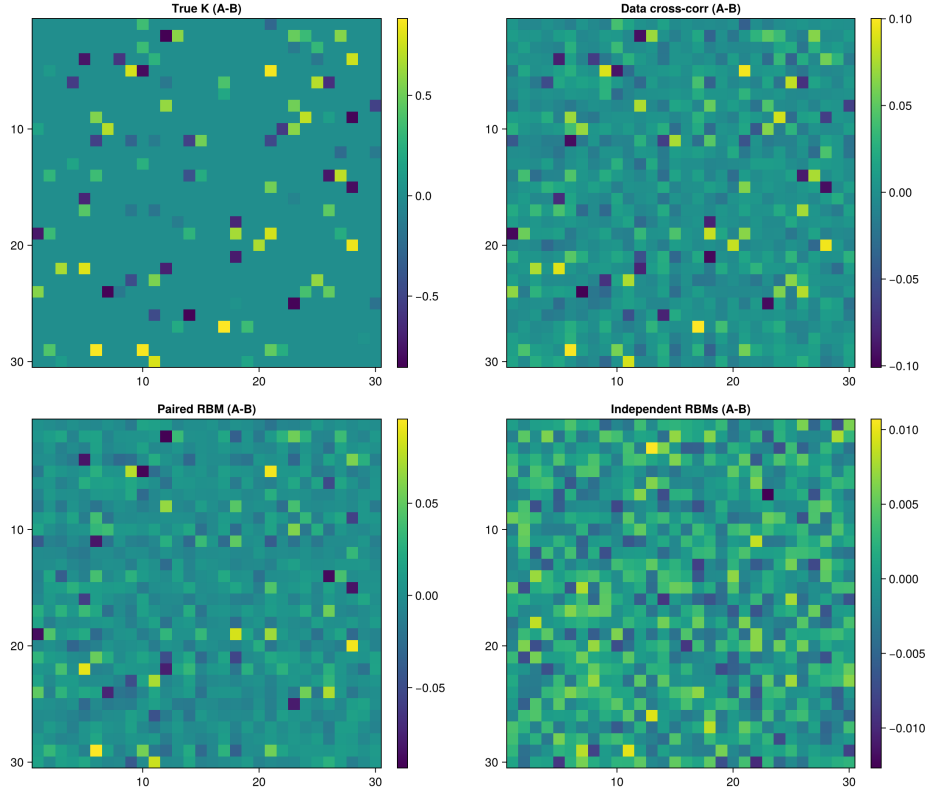


Figure 10: K -recovery heatmap: true K , empirical data cross-correlation, paired-model recovery, independent-RBM baseline.

Pfam: PF00072–PF00512, $K = 50$

$N_{\text{hidden},A} = 150$, $N_{\text{hidden},B} = 100$, $H_{\text{add}} = 20$, $N_{\text{iters}} = 10000$, $\text{paired_iters} = 5000$. All figures produced for this run.

Training analysis

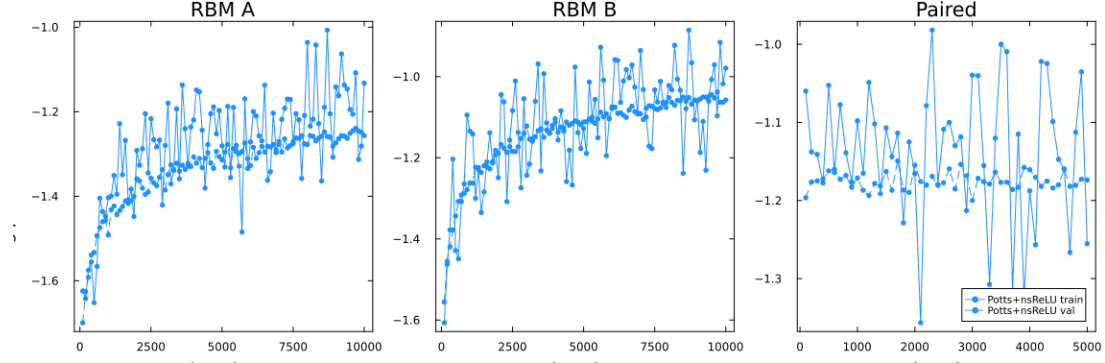


Figure 11: Pseudolikelihood, train/validation, private RBMs A (PF00072), B (PF00512), and paired model.

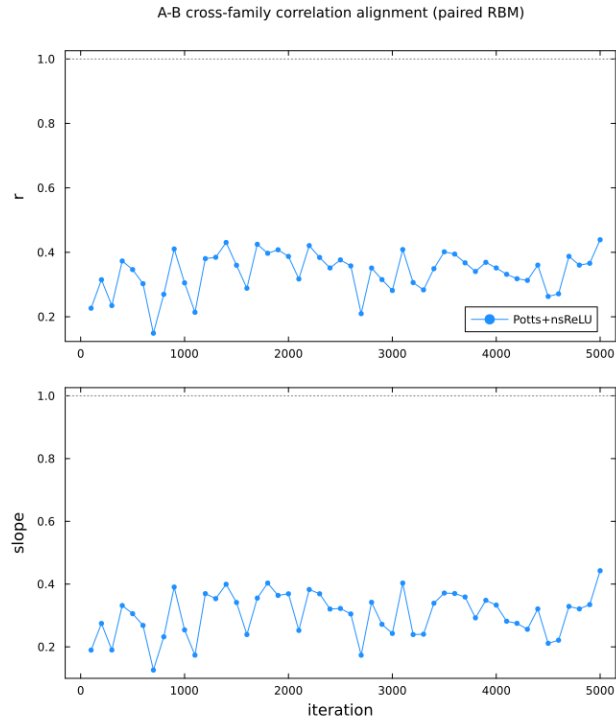


Figure 12: Training-time interfamily (ab -check) correlation.

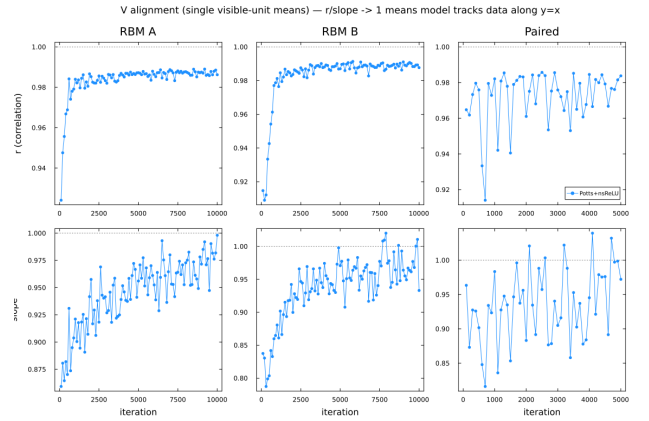


Figure 13: Single-site visible marginals $\langle v \rangle$, data vs. model.

Pair results (equilibrium Gibbs)

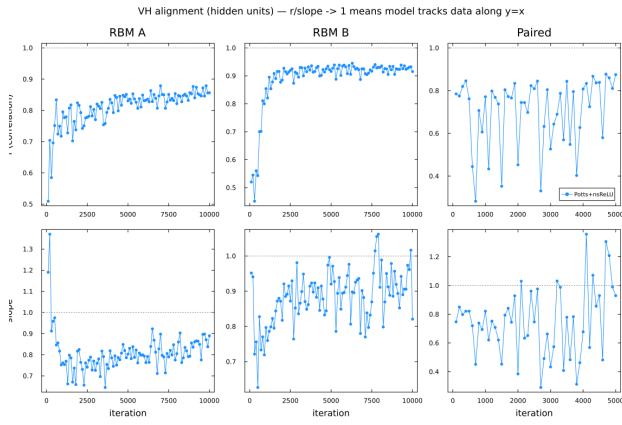


Figure 14: Joint $\langle vh \rangle$, added units, data vs. model.

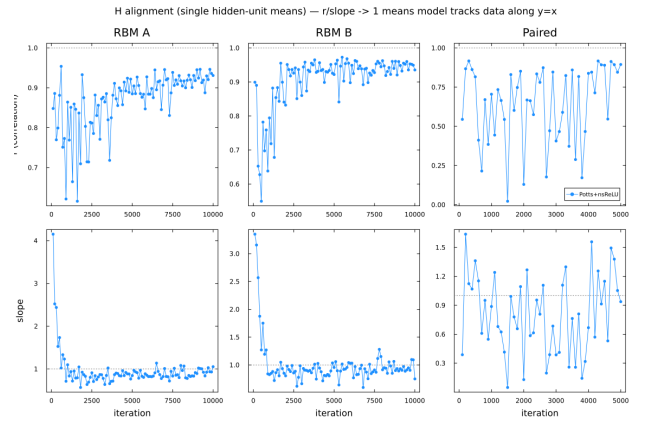


Figure 15: Single hidden-unit means $\langle h \rangle$, data vs. model.

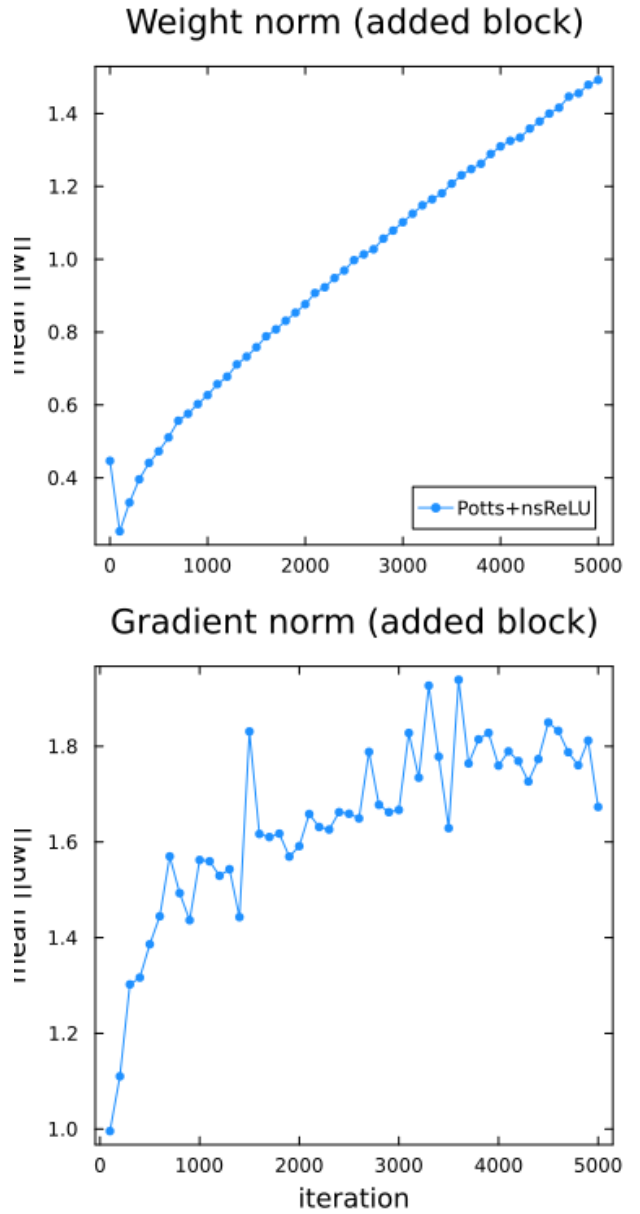


Figure 17: Weight-norm/gradient-norm of the added block over training.

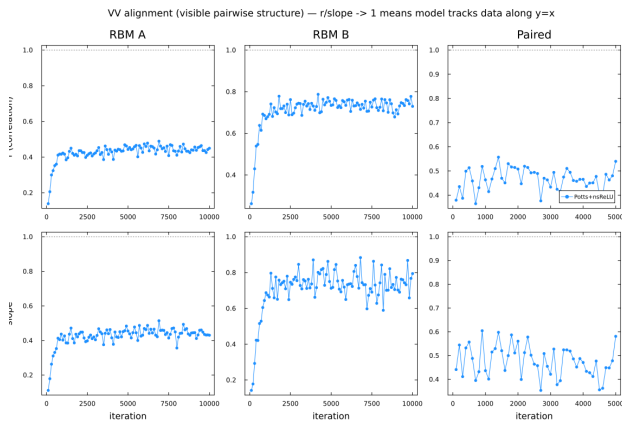


Figure 16: Pairwise visible-visible correlation, private RBMs, data vs. model.

ring rate at last checkpoint (near $y=x$ & away f
Potts+nsReLU

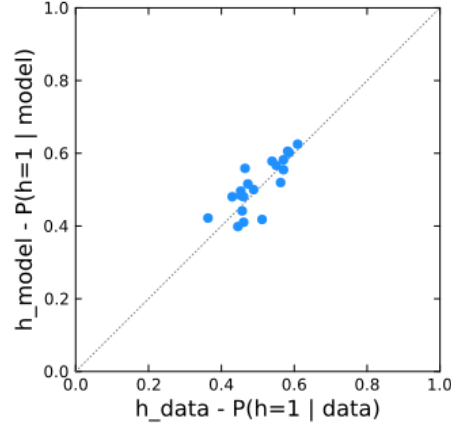


Figure 18: Added-unit firing rate, data vs. model (dead-unit check).

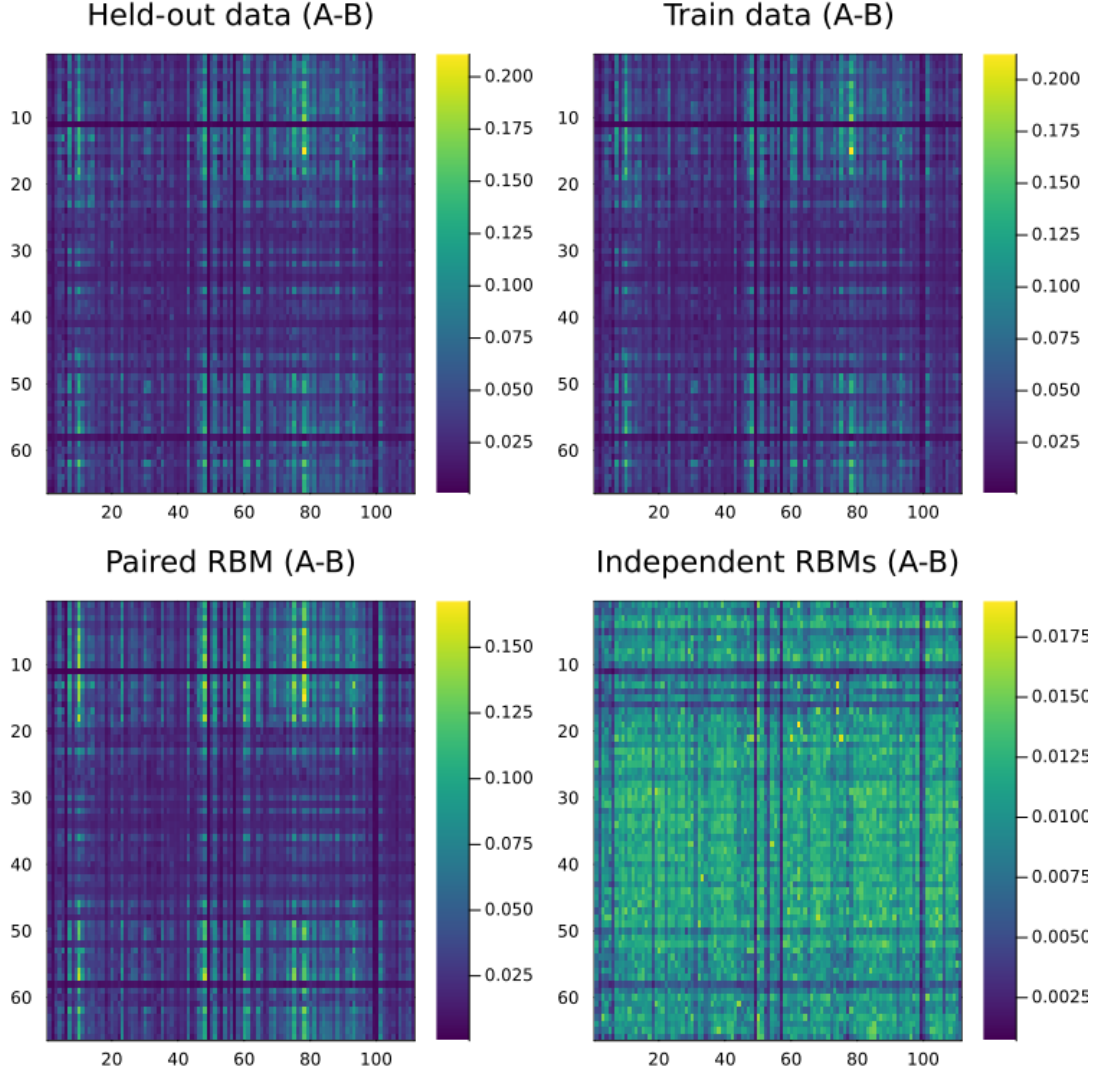


Figure 19: Decisive test: cross-family coupling-strength heatmaps, held-out data, training data, paired model, independent-RBM baseline. Top-366 overlap = 0.814 (ceiling 0.956, null 0.044); whole-matrix correlation = 0.949 (ceiling 0.997, null 0.178).

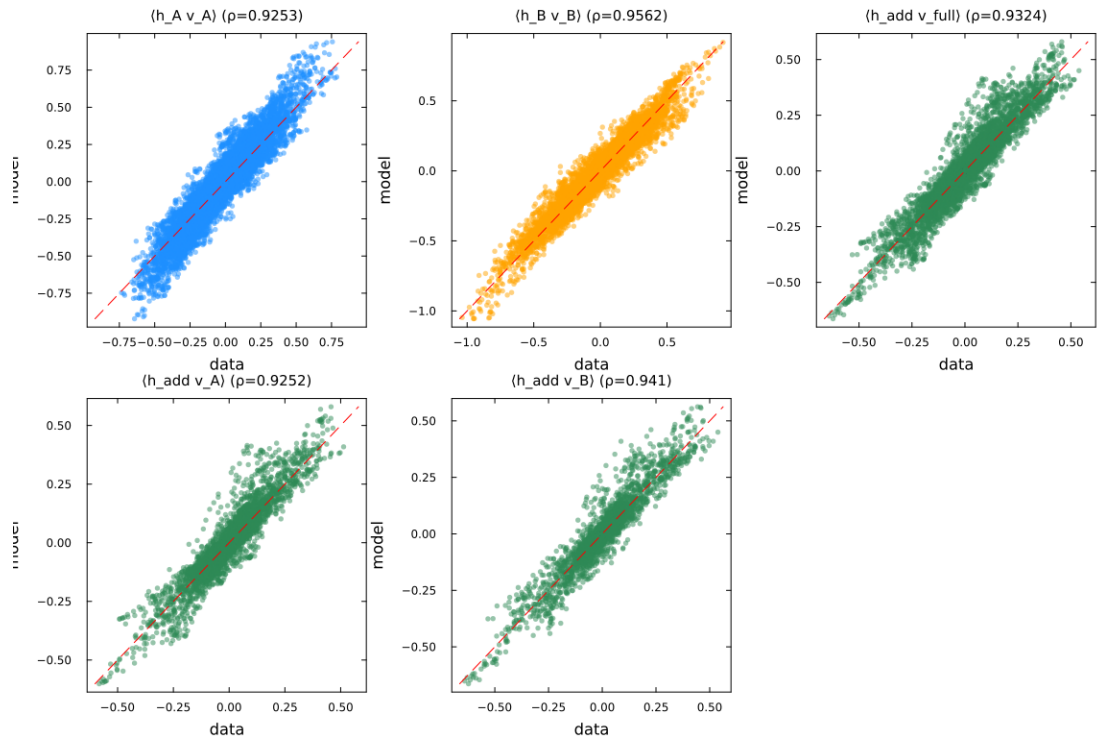


Figure 20: Block-wise $\langle hv \rangle$ at equilibrium: frozen A - A , frozen B - B , added units vs. full visible layer.

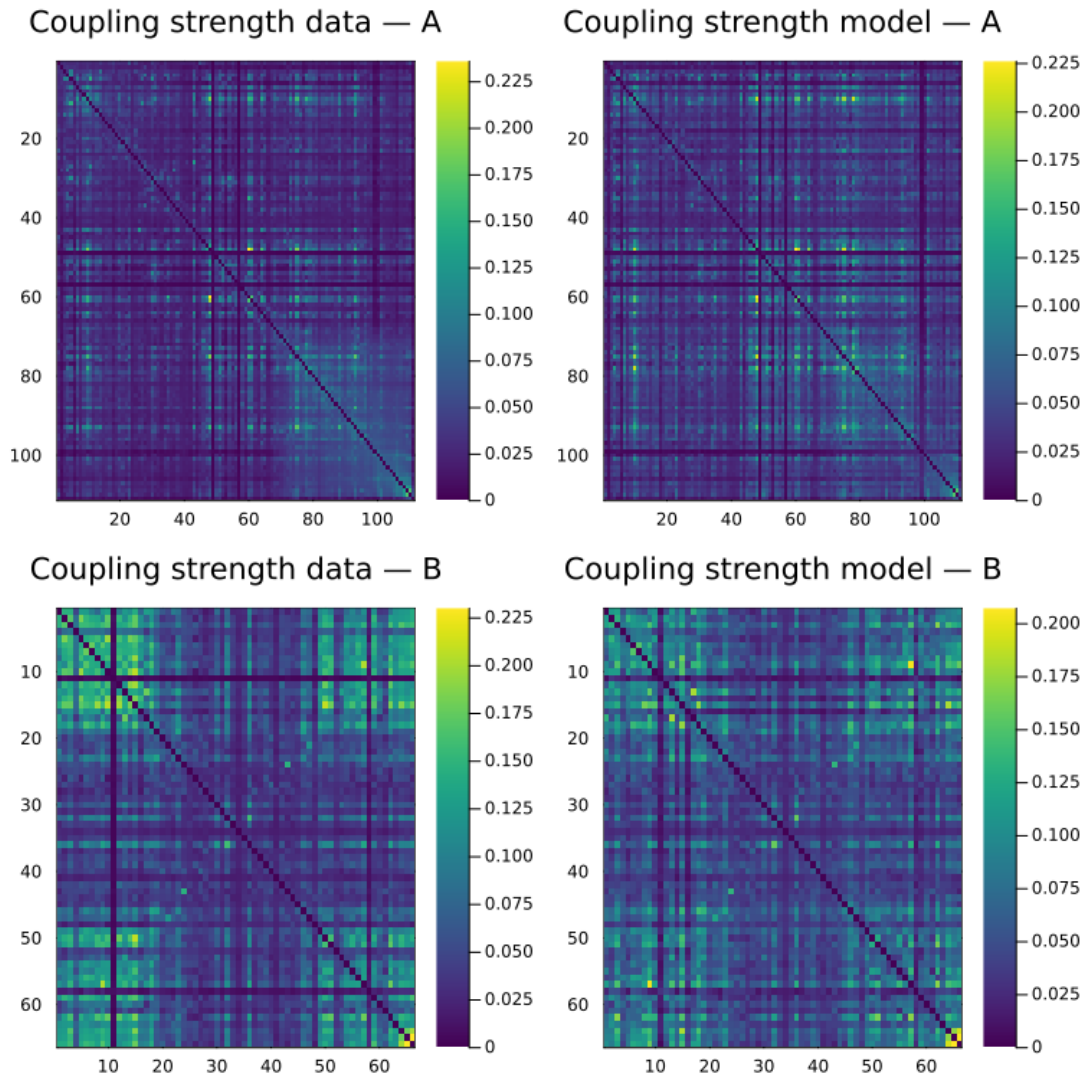


Figure 21: Per-family coupling-strength heatmaps, data vs. model, at equilibrium.

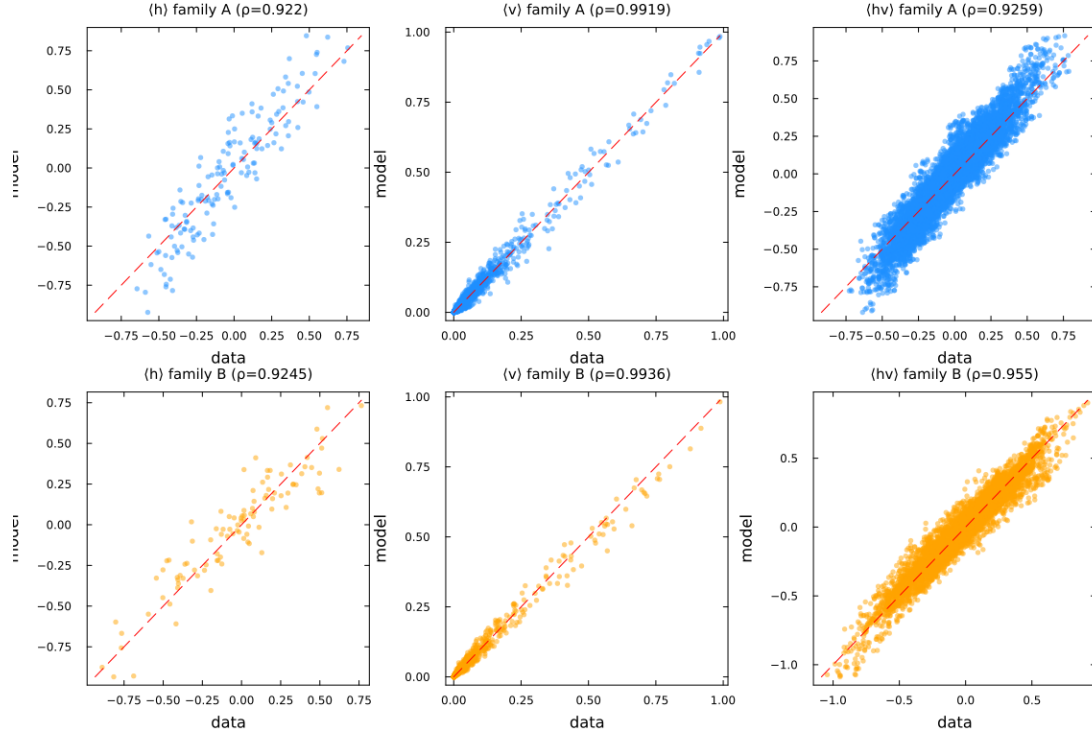


Figure 22: Intra-family $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$ at equilibrium, families A (top) / B (bottom). Mean off-diagonal coupling data $\rightarrow$ paired model: A $0.035 \rightarrow 0.038$, B $0.061 \rightarrow 0.070$.

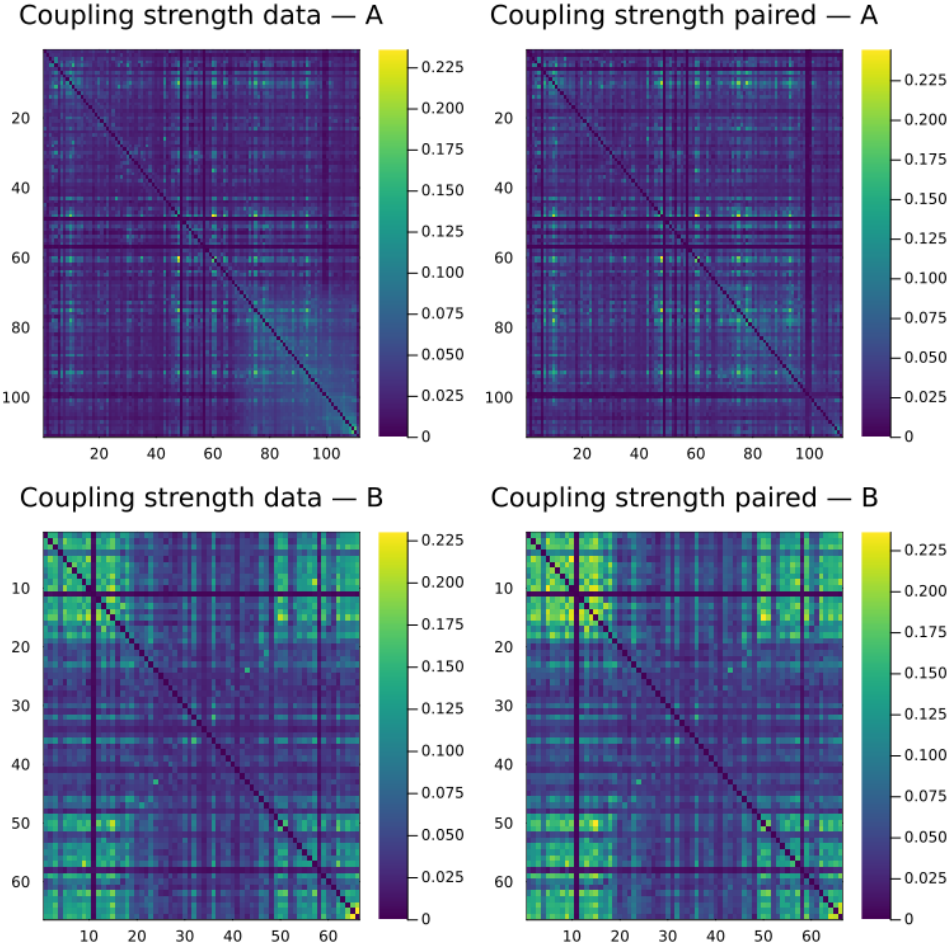


Figure 23: Intra-family coupling-strength heatmaps, data vs. paired model, families A/B .

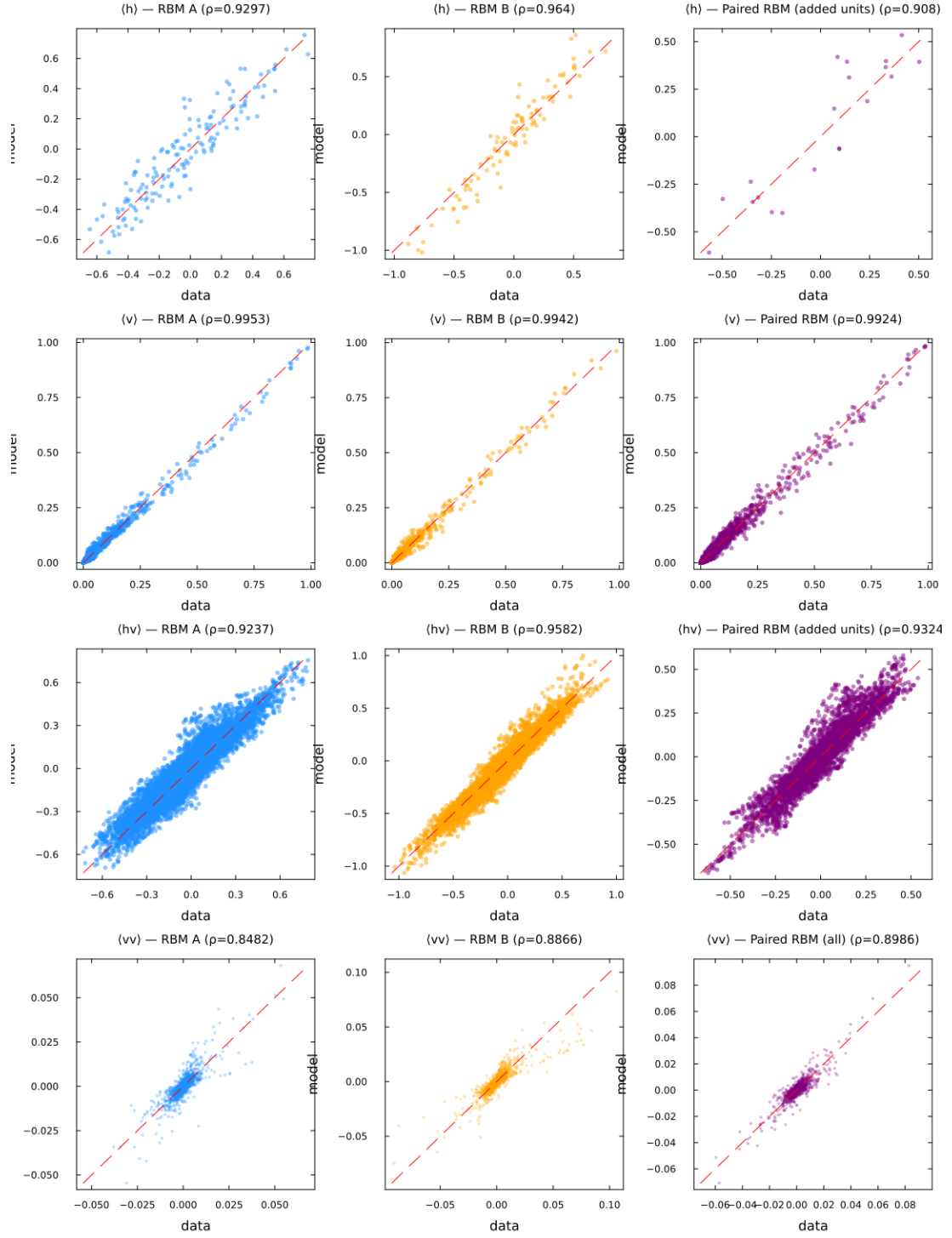


Figure 24: Full moment validation at equilibrium ($\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$, $\langle vv \rangle$), private RBMs and paired model.

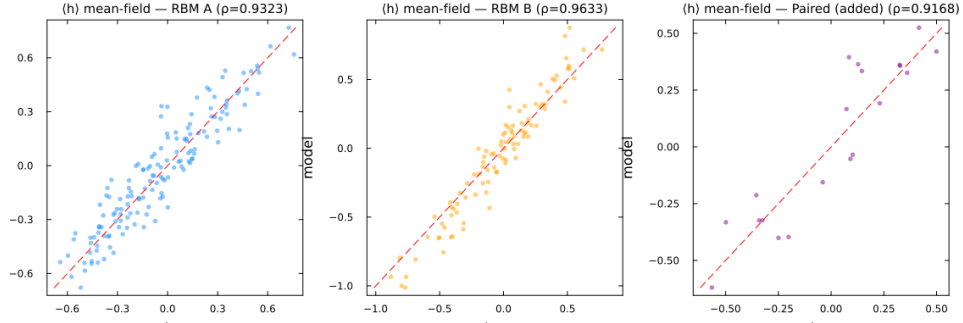


Figure 25: Hidden-unit mean-field consistency check, RBM A , B , and paired (added units).

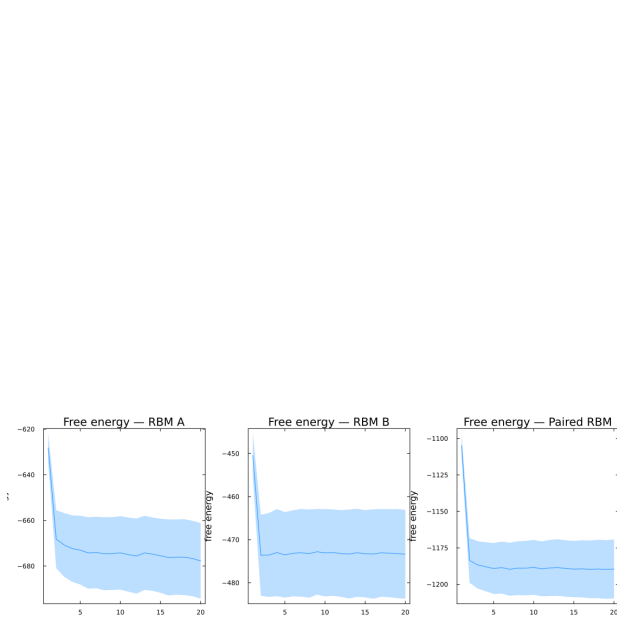


Figure 26: Free energy vs. Gibbs sweep count – equilibration/burn-in check.

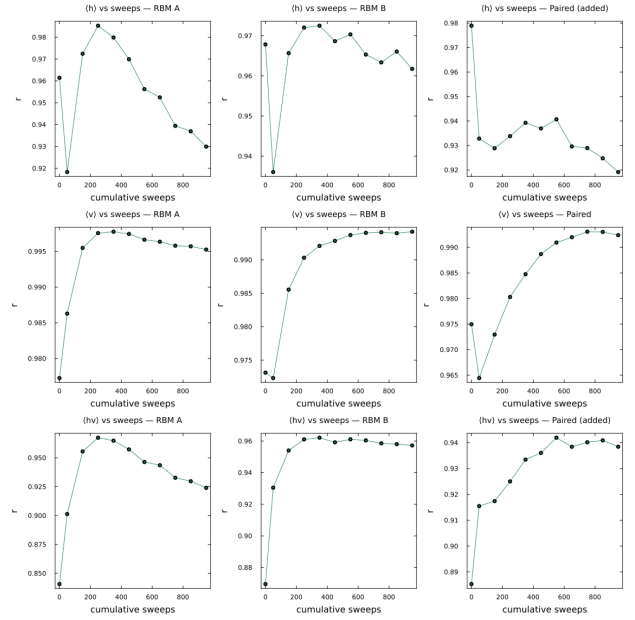


Figure 27: Convergence of $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$ correlations vs. cumulative sweeps, RBM A , B , paired.

Pfam: PF00072–PF01339

$N_{\text{hidden},A} = 150$, $N_{\text{hidden},B} = 100$, $H_{\text{add}} = 20$, $N_{\text{iters}} = 15000$, $\text{paired_iters} = 8000$. All figures produced for this run.

Training analysis

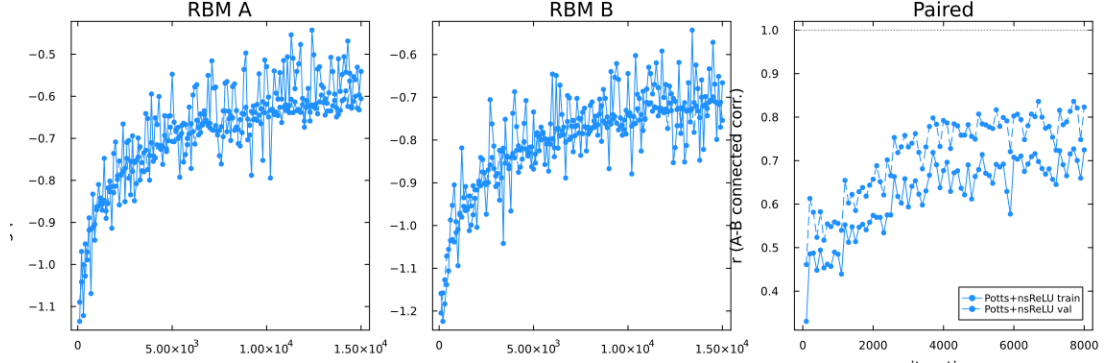


Figure 28: Pseudolikelihood, train/validation, private RBMs A (PF00072), B (PF01339), and paired model.

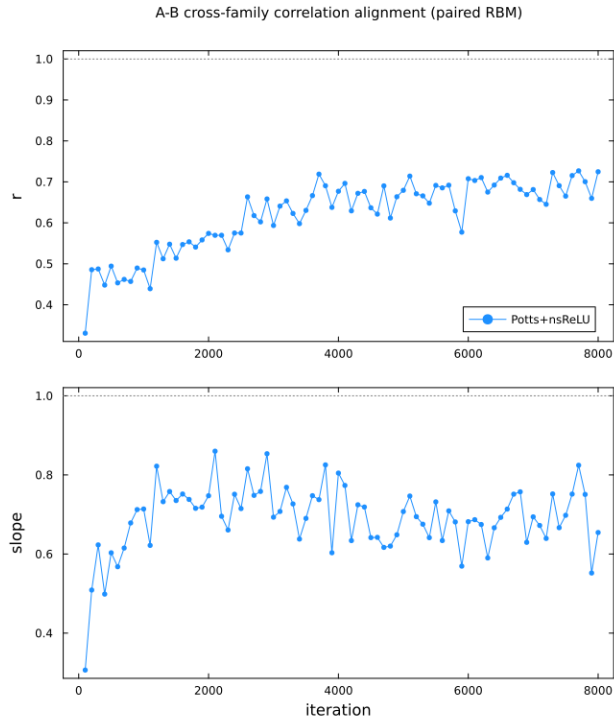


Figure 29: Training-time interfamily (ab -check) correlation: $r = 0.69$ train / 0.79 held-out.

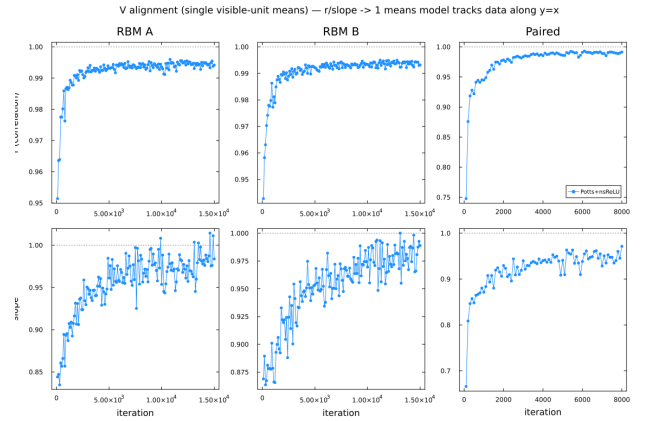


Figure 30: Single-site visible marginals $\langle v \rangle$, data vs. model.

Pair results (equilibrium Gibbs)

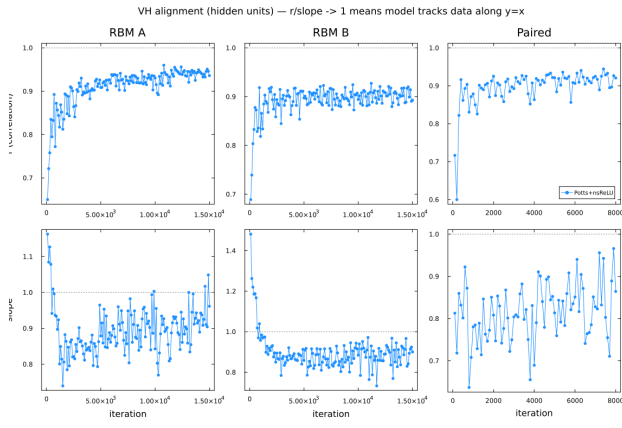


Figure 31: Joint $\langle vh \rangle$, added units, data vs. model.

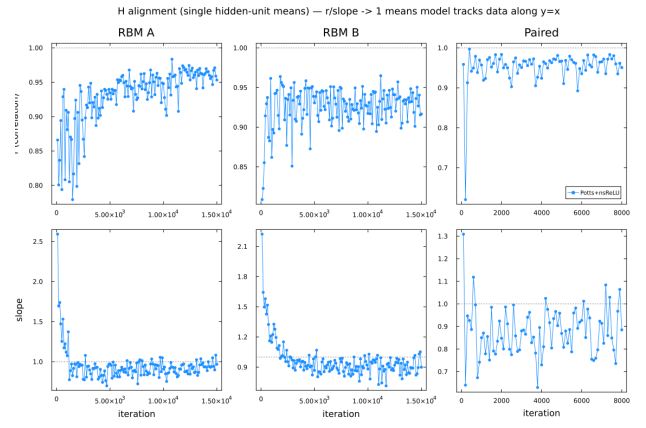


Figure 32: Single hidden-unit means $\langle h \rangle$, data vs. model.

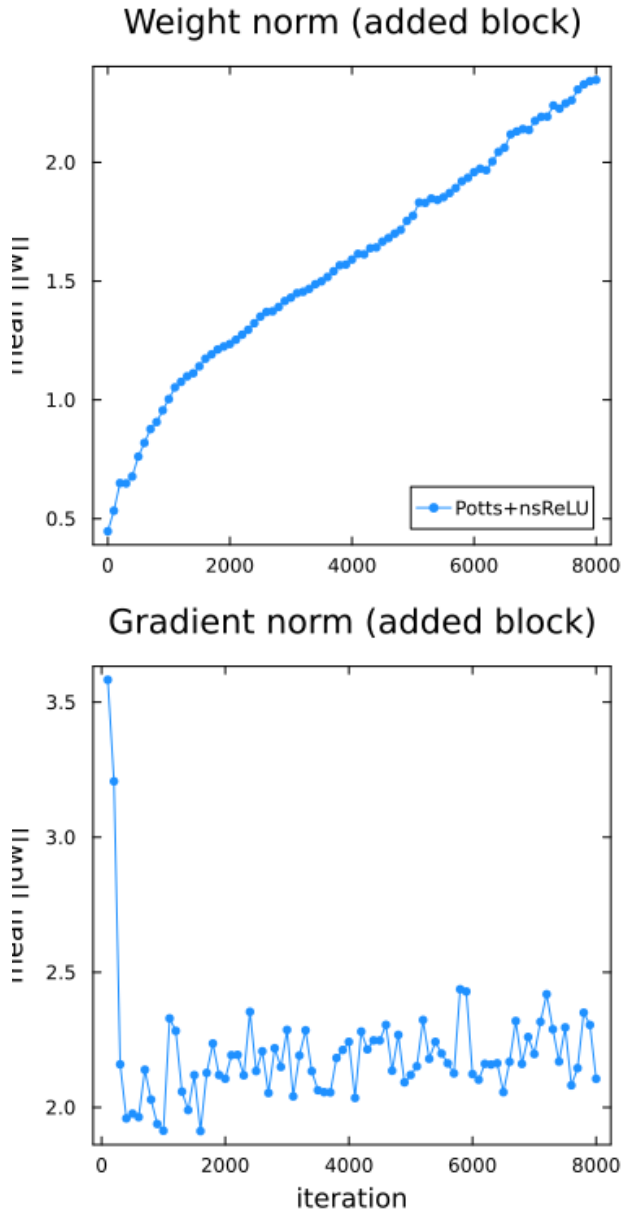


Figure 33: Weight-norm/gradient-norm of the added block over training.

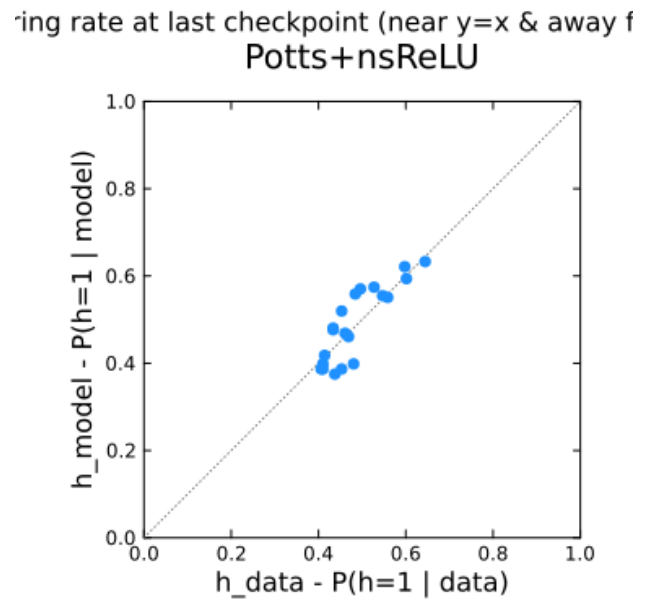


Figure 34: Added-unit firing rate, data vs. model (dead-unit check).

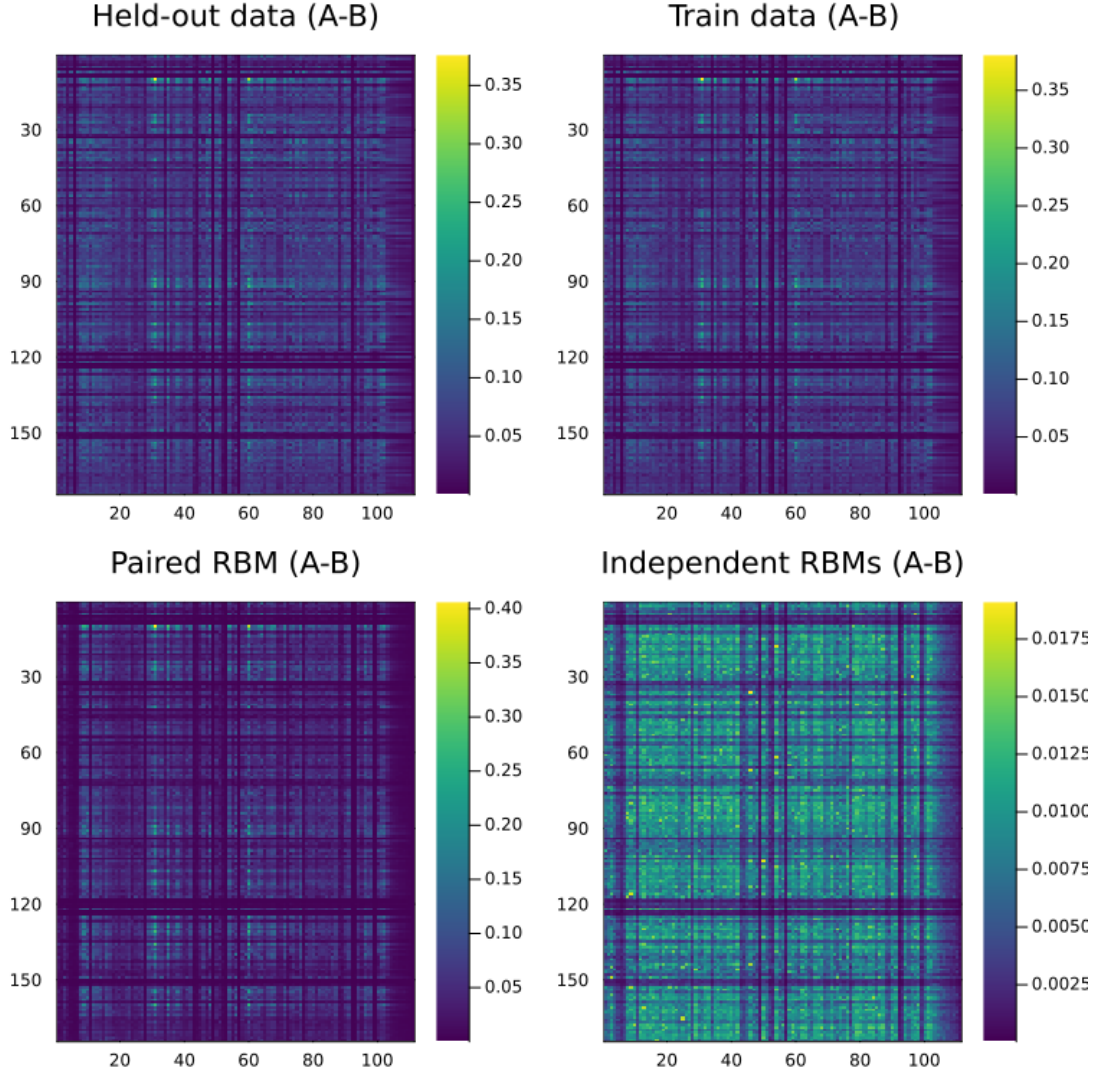


Figure 35: Decisive test: cross-family coupling-strength heatmaps, held-out data, training data, paired model, independent-RBM baseline. Top-966 overlap = 0.509 (ceiling 0.942, null 0.045); whole-matrix correlation = 0.733 (ceiling 0.996, null 0.539 – null floor much higher here than for PF00512, likely shared conservation pattern rather than genuine coupling).

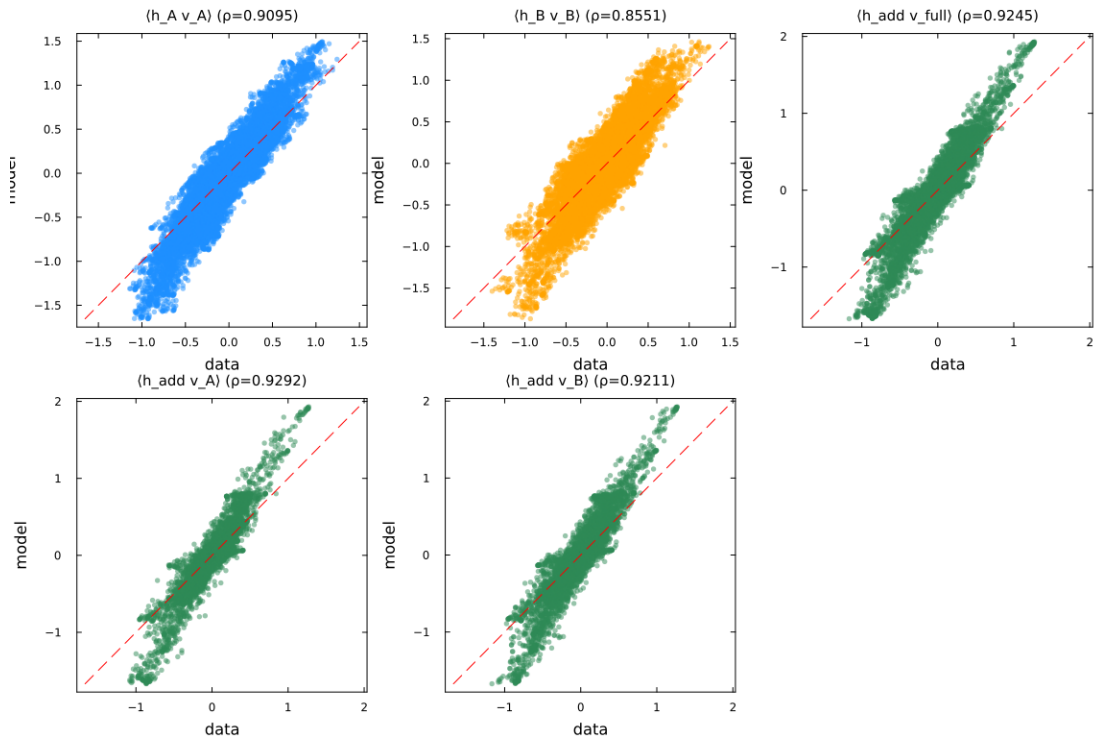


Figure 36: Block-wise $\langle hv \rangle$ at equilibrium: frozen A - A , frozen B - B , added units vs. full visible layer.

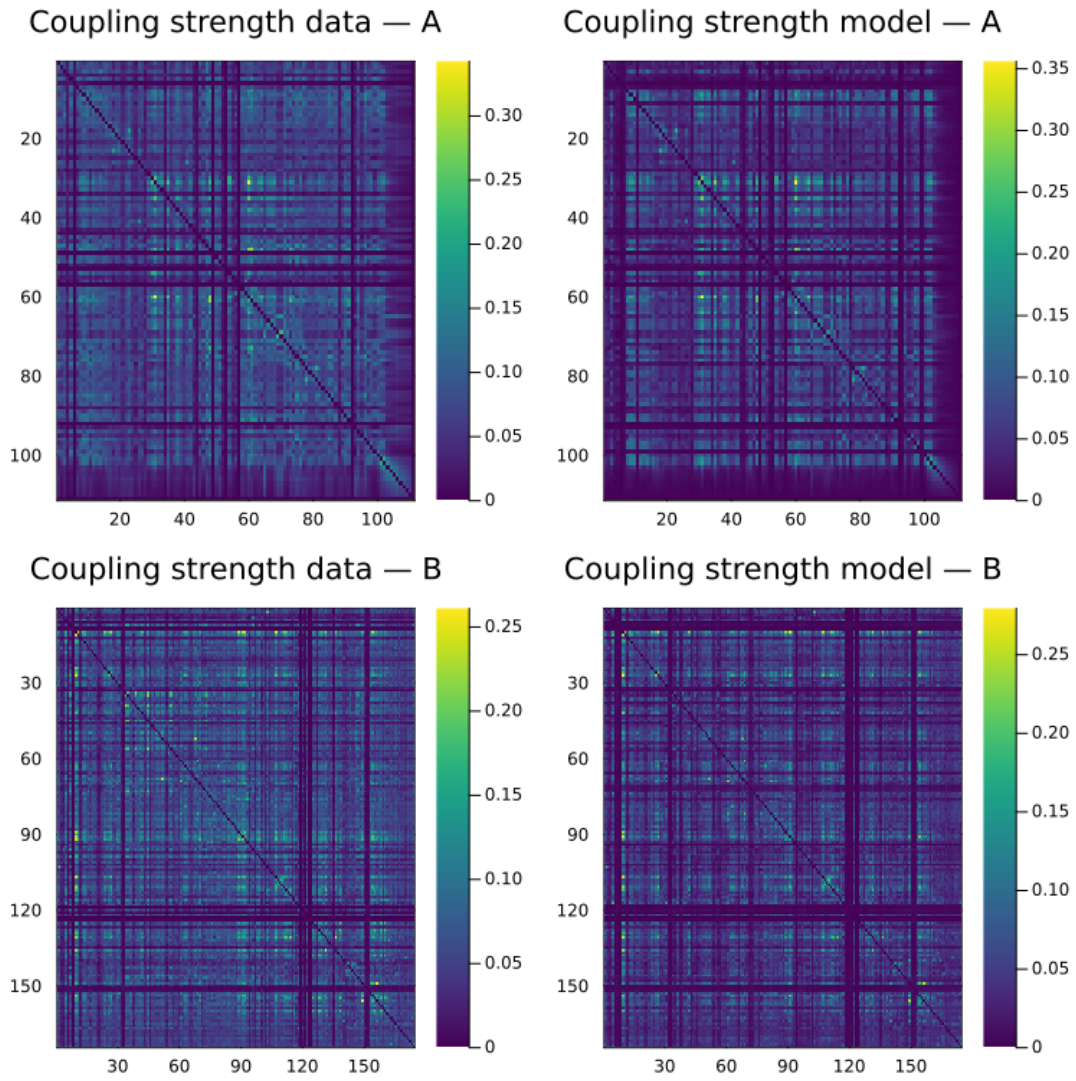


Figure 37: Per-family coupling-strength heatmaps, data vs. model, at equilibrium.

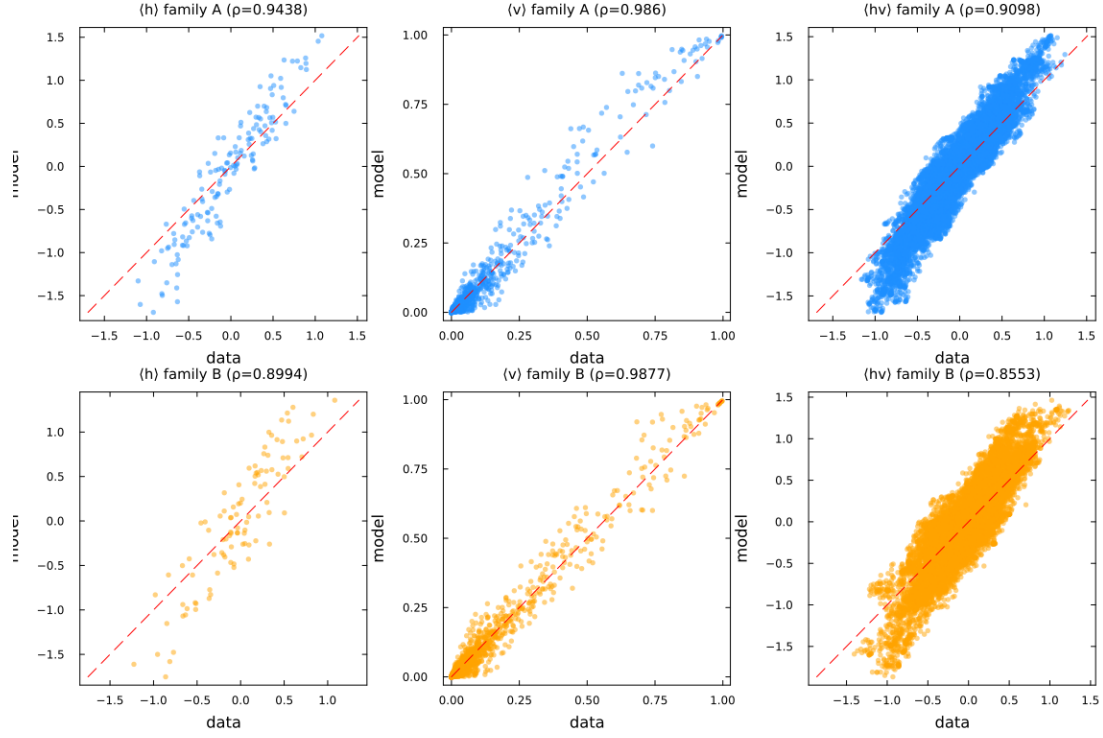


Figure 38: Intra-family $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$ at equilibrium, families A (top) / B (bottom). Mean off-diagonal coupling data $\rightarrow$ paired model: A $0.056 \rightarrow 0.039$, B $0.049 \rightarrow 0.041$.

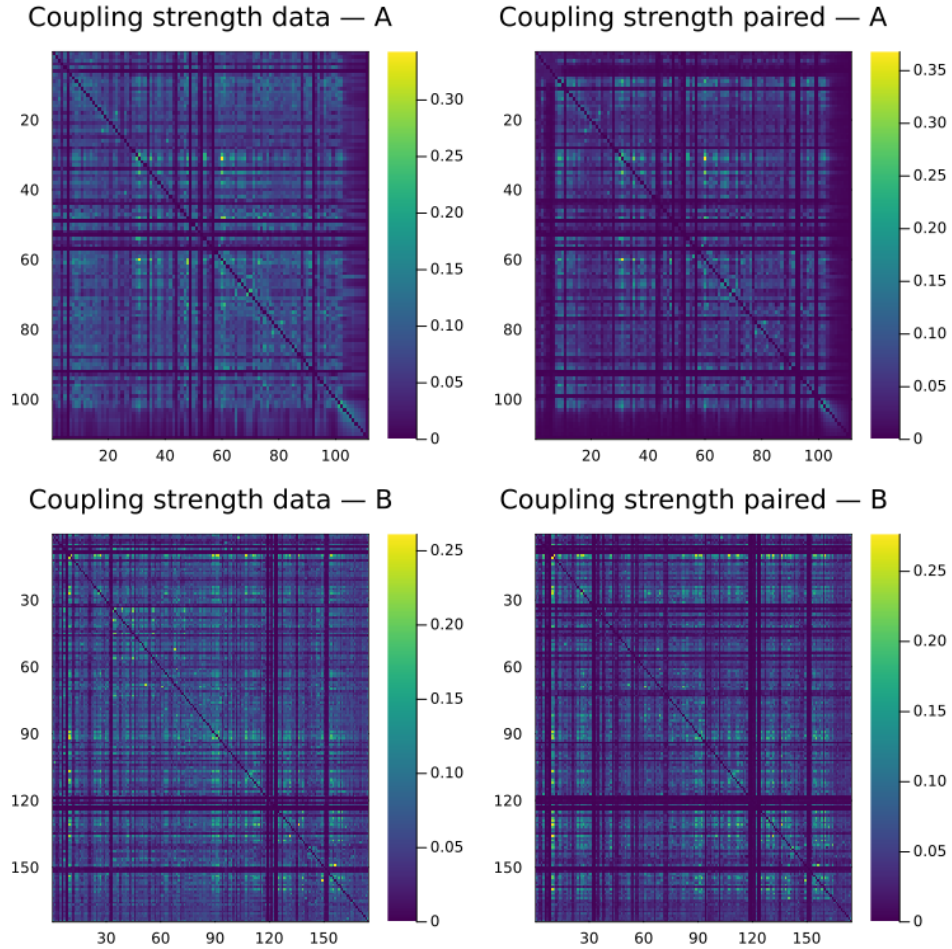


Figure 39: Intra-family coupling-strength heatmaps, data vs. paired model, families A/B .

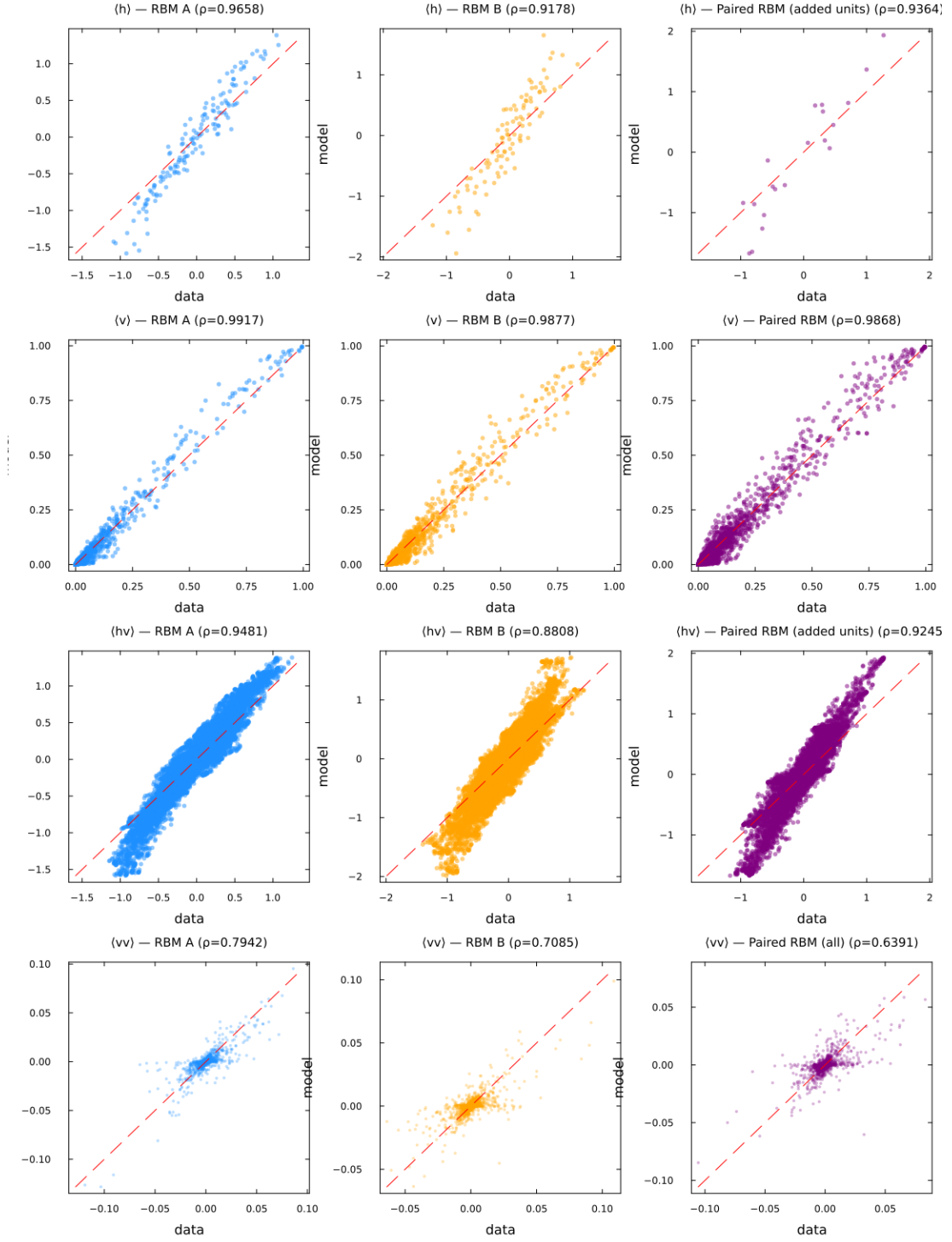


Figure 40: Full moment validation at equilibrium ($\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$, $\langle vv \rangle$), private RBMs and paired model.

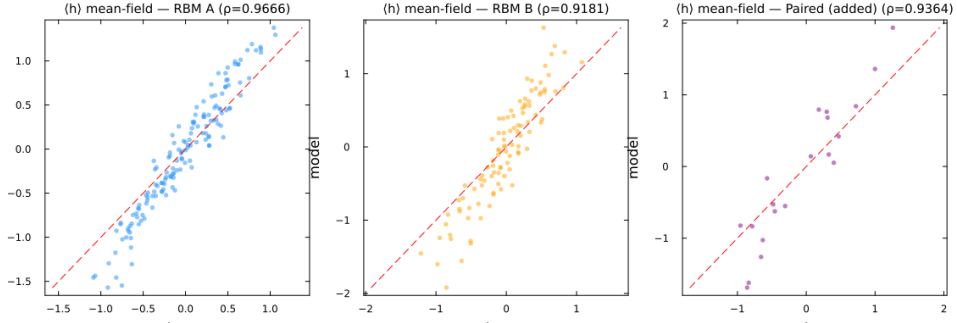


Figure 41: Hidden-unit mean-field consistency check, RBM A , B , and paired (added units).

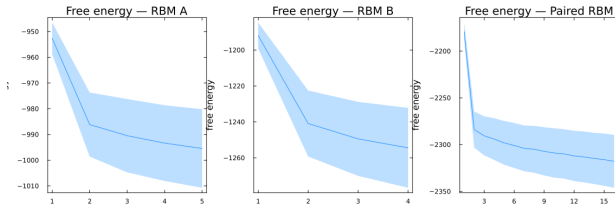


Figure 42: Free energy vs. Gibbs sweep count – equilibration/burn-in check.

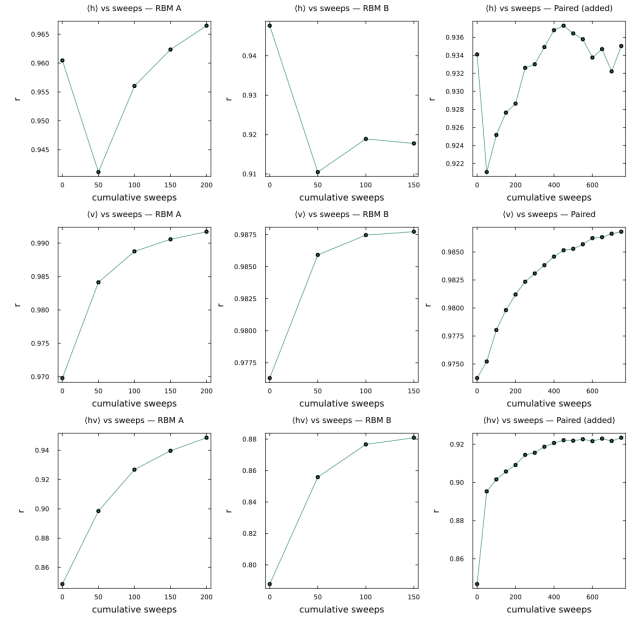


Figure 43: Convergence of $\langle h \rangle$, $\langle v \rangle$, $\langle hv \rangle$ correlations vs. cumulative sweeps, RBM A , B , paired.